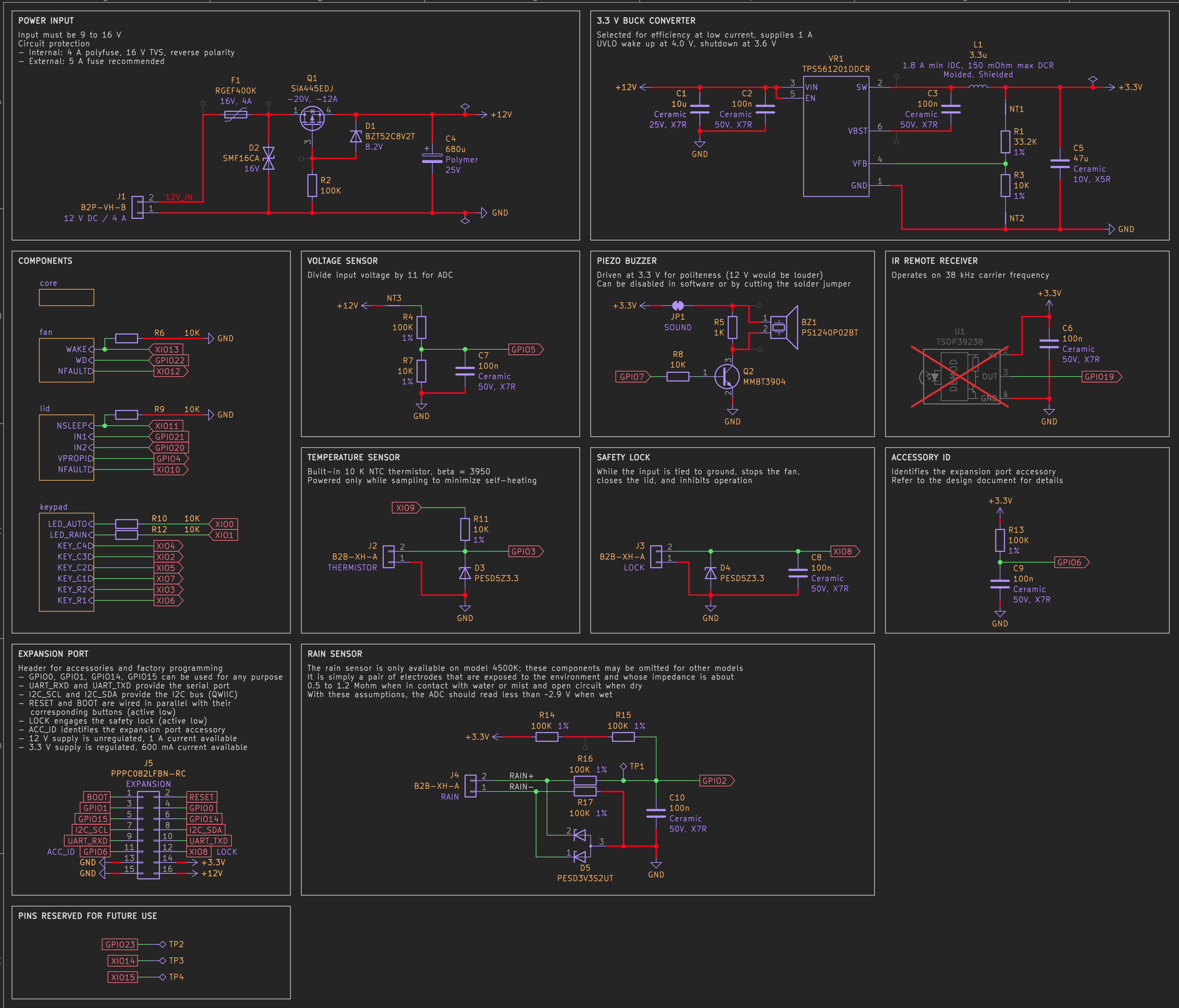
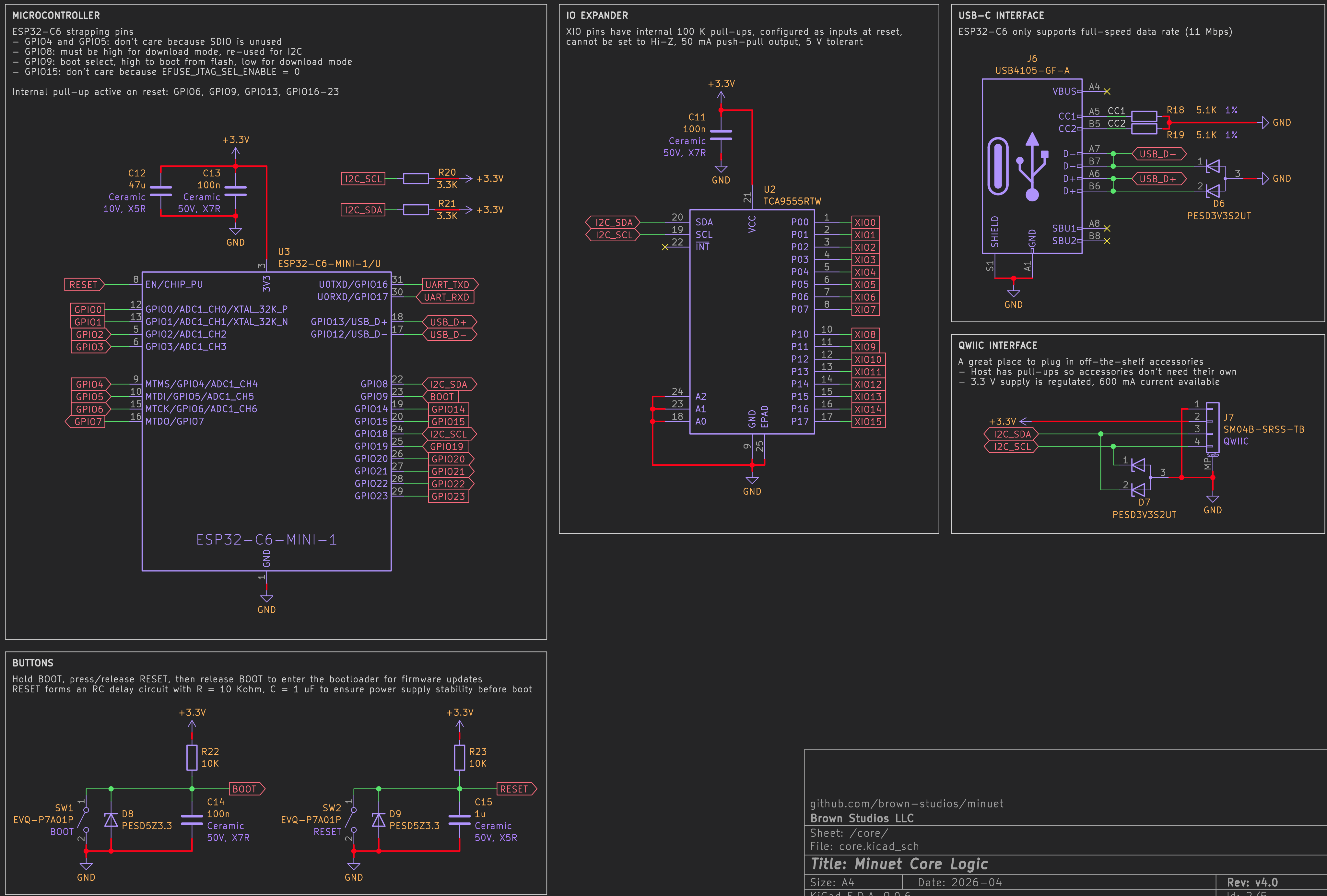


github.com/brown−studios/minuet
Brown Studios LLC
Sheet: /
File: minuet.kicad_sch
Title: Minuet Fan Controller

Size: A3	Date: 2026−04	**Rev: v4.0**
KiCad E.D.A. 9.0.6		Id: 1/5



github.com/brown–studios/minuet
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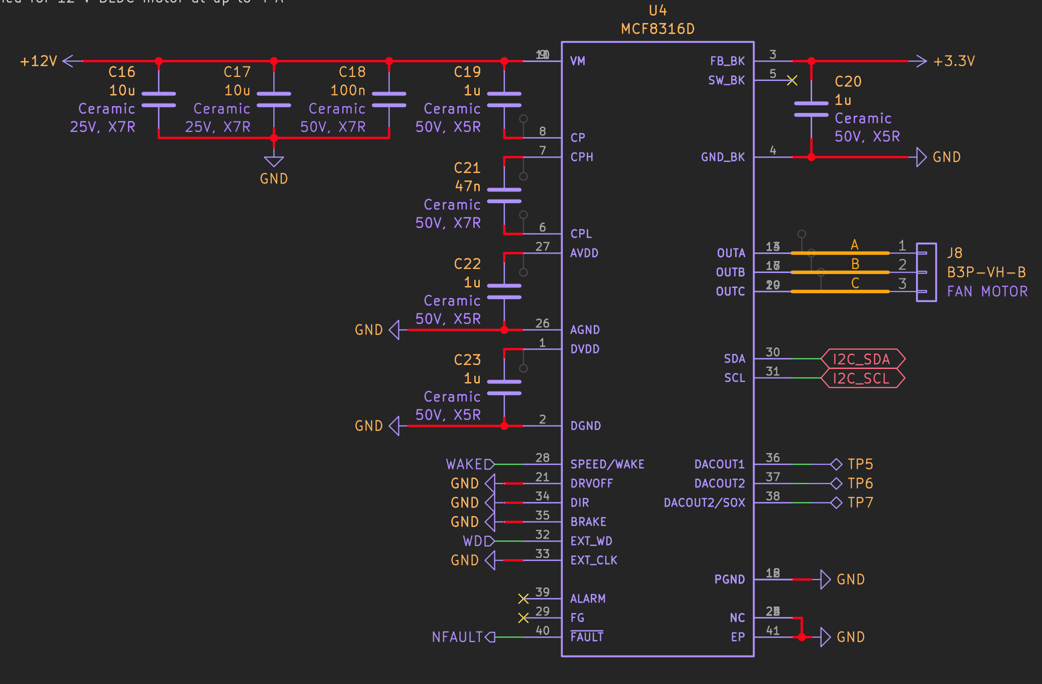
Sheet: /core/
File: core.kicad_sch

Title: Minuet Core Logic

Size: A4	Date: 2026–04	**Rev: v4.0**
KiCad E.D.A. 9.0.6		Id: 2/5

FAN MOTOR DRIVER

Designed for 12 V BLDC motor at up to 4 A



AVDD supplies 3.3 V for internal analog circuitry from VM when not sleeping
DVDD supplies 1.5 V for internal digital logic from FB_BK when not sleeping
FB_BK is externally powered with 3.3 V and the buck converter is disabled to reduce the BOM per application note SLLA643

NFAULT is open-drain, has an internal pull-up to AVDD that can be disabled via I2C, SLLA643 says never use external and internal pull-up together
SPEED/WAKE has an internal 1 Mohm pull-down for noise immunity and is configured via I2C to wake the device from sleep when high
I2C controls all device functions except wake-up and watchdog

github.com/brown-studios/minuet
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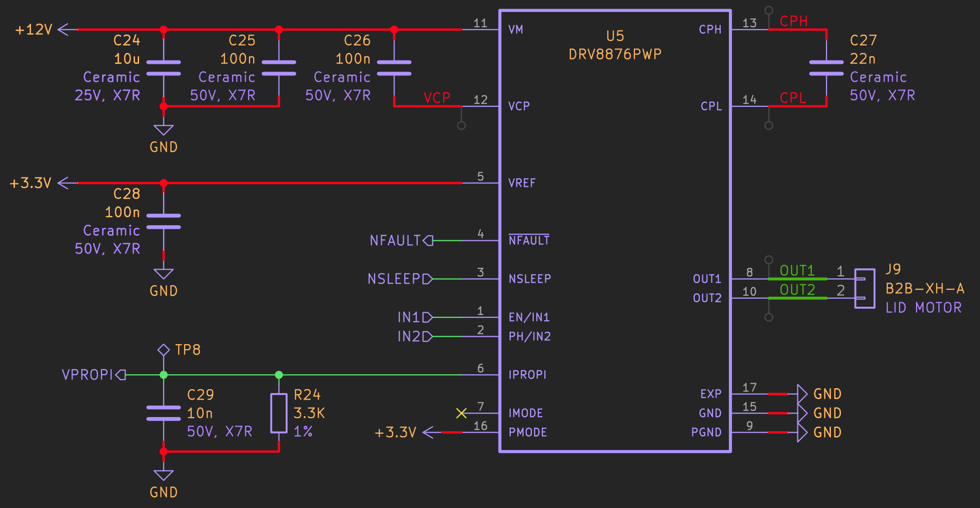
Sheet: /fan/
File: fan.kicad_sch

Title: Minuet Fan Motor Driver

Size: A4	Date: 2026-04	**Rev: v4.0**
KiCad E.D.A. 9.0.6		Id: 3/5

LID MOTOR DRIVER

Designed for 12 V DC motor at up to 2 A



PMODE set to PWM mode with IN1 / IN2 signals
IMODE set to mode 4: fixed off time current chopping, overcurrent latches output off and reports a fault
Itrip (A) = 1000 * Vref (V) / Rpropi (Ω) = 1 A
NSLEEP activates motor when high
NFAULT is open-drain

Internal pull-down on NSLEEP, IN1, IN2

github.com/brown-studios/minuet
Brown Studios LLC
Sheet: /lid/
File: lid.kicad_sch
Title: Minuet Lid Motor Driver

Size: A4	Date: 2026-04	**Rev: v4.0**
KiCad E.D.A. 9.0.6		Id: 4/5

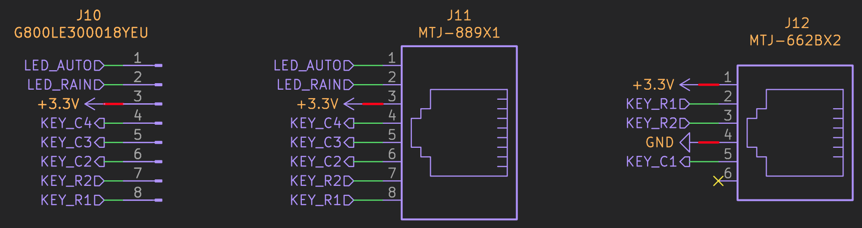
KEYPAD AND WIRED WALL CONTROL INTERFACE

Compatible with original equipment

Key matrix
- R1+GND: Close lid
- R2+GND: Open lid
- R1+C1: Fan off
- R1+C2: Fan speed up
- R1+C3: Fan speed down
- R1+C2+C3: Open / close lid
- R1+C4: Toggle rain sensor (only certain models)
- R2+C1: Fan on and cycle speeds
- R2+C2: Fan on / off
- R2+C3: Fan intake / exhaust direction
- R2+C4: Toggle auto thermostat

LEDs
- LED_AUTO: Auto LED cathode, green with 2.4 V voltage drop
- LED_RAIN: Rain LED cathode, red with 2 V voltage drop (only certain models)

KEY_Rx and KEY_Cx inputs need pull-ups
Detect key press when KEY_Rx goes low when undriven (tristate)
Detect key press when KEY_Cx goes low during high-to-low strobe of KEY_Rx
Fan originally provided +5V supply the interface but +3.3V works fine



A note on 8P8C (RJ45) and 6P6C (RJ12/RJ11) jacks:

These jacks are available in numerous footprints and form factors. For this application, we prefer a jack that is sealed on the top to keep out condensation and a tab on the bottom for ease of access such as MTJ-889X1 and MTJ-669X1. However, while these specific parts are readily available at Digikey, they are are not normally stocked and relatively expensive at JLCPCB so we must also consider alternatives for manufacturing.

Selected equivalent parts:

8P8C (tab down, 17.3 mm OAL)
- Adam Tech MTJ-889X1
- Amphenol 54602-908LF
- HCTL HC-RJ45-058-1-6
- XunPu RJ45-306-005-18
- XKB X31ADIWA3DY1027

6P6C (tab up, 13 mm OAL):
- Adam Tech MTJ-662BX1/MTJ-662BX2
- CONNFLY Elec DS1133-S60BPX
- Ckmtw R-RJ11R06P-A800

github.com/brown-studios/minuet **Brown Studios LLC**		
Sheet: /keypad/ File: keypad.kicad_sch		
Title: Minuet Keypad Interface		
Size: A4	Date: 2026-04	**Rev: v4.0**
KiCad E.D.A. 9.0.6		Id: 5/5